

Housing Price and Rent Prediction

Final Exam: Quant Modeling

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2023 Dual Bachelor's Degree Program in Econ and Math

Data Processing

清洗：统一数值格式，Winsorize(5%/95%)处理异常

数值编码：分类特征转为哑变量

列删减：移除无关冗余列

创新点1：HDBSCAN 地理聚类

→ 其他条件不变时，kaggle分数由63.9提升至71.9

Data Processing

创新点2: Knowledge Distillation(小样本蒸馏)

随机抽取300条数据用LLM读取

TF-IDF+逻辑回归训练模型，高效预测剩余数据

Column	Positive	Negative	Neutral
核心卖点	80.66%	0.00%	19.34%
户型介绍	28.13%	0.00%	71.87%
周边配套	53.61%	0.00%	46.39%
交通出行	44.84%	0.00%	55.16%
客户反馈_Price	44.59%	54.96%	0.45%
客户反馈_Rent	53.44%	44.93%	1.63%

Model Performances

[Price Dataset Metrics Table]			
Model	In sample rmse	out of sample rmse	Cross-validation rmse
XGB	3.917399e+05	4.639423e+05	4.849306e+05
Blended	4.078968e+05	4.852599e+05	5.344251e+05
LGBM	5.440214e+05	5.689826e+05	5.784991e+05
RF	4.405285e+05	5.573310e+05	5.799463e+05
Cat	6.257903e+05	6.363408e+05	6.521011e+05
Torch	4.119864e+05	6.584406e+05	7.974345e+05
Keras	2.504335e+07	1.179186e+07	1.670565e+07

[Rent Dataset Metrics Table]			
Model	In sample rmse	out of sample rmse	Cross-validation rmse
XGB	106669.341886	124075.584012	1.314175e+05
Blended	105261.743907	124916.502409	1.360594e+05
RF	96554.335183	130588.160326	1.385385e+05
LGBM	133945.840250	137465.024110	1.433259e+05
Cat	148375.502563	148390.496052	1.550775e+05
Torch	131412.053186	154859.102680	1.571160e+05
Keras	196481.780632	194249.643660	3.355455e+10

最终提交文件前5行:

	ID	Price
0	1000000	14357159.00
1	1000001	3210549.50
2	1000002	5104397.00
3	1000003	2426900.75
4	1000004	9585259.00

最终提交文件前5行:

	ID	Price
0	1000000	13843006.00
1	1000001	3311418.00
2	1000002	4692432.00
3	1000003	2430254.25
4	1000004	10141600.00

Model	In	OOS	CV	KS
XGB_price	391.7k	463.9k	484.9k	75.9
XGB_rent	106.6k	124.0k	131.4k	75.9
XGB_price*	392.9k	464.7k	485.8k	75.9
XGB_rent*	106.3k	124.8k	131.6k	75.9

In = In-sample RMSE
OOS = Out-of-sample RMSE
CV = Cross-validation RMSE
KS = Kaggle Score
All values in thousands(k)